

Project Presentation of CO-OP Project(Module-2) (22CS421)
On

An Explainable ML Framework for Binary Classification

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Objectives

Motivate

Build an explainable ML framework to bring transparency to binary classification decisions in critical domains.

Study

Evaluate SHAP and LIME for explanation consistency, faithfulness, and stability across multiple re-training runs.

Apply

Benchmark four classifiers (LR, RF, XGBoost, SVM) on three real-world datasets to compare accuracy and interpretability.

Explore

Deliver a modular, domain-independent XAI pipeline that practitioners can deploy for responsible AI in the real world.

Introduction

- ML models are applied across critical
 - domains: healthcare diagnostics,
 - financial risk scoring, cybersecurity threat detection, and HR analytics.
- Despite strong predictive performance, complex models like XGBoost and neural networks remain opaque black boxes.
- Post-hoc explainability tools SHAP and LIME bridge the gap between model performance and human understanding.
- Explainability is essential for regulatory compliance and building trust in AI-driven decisions.

Introduction

SHAP Method

- Merit-based Ranking
- Higher Trust
- SEO Dependent

LIME Method

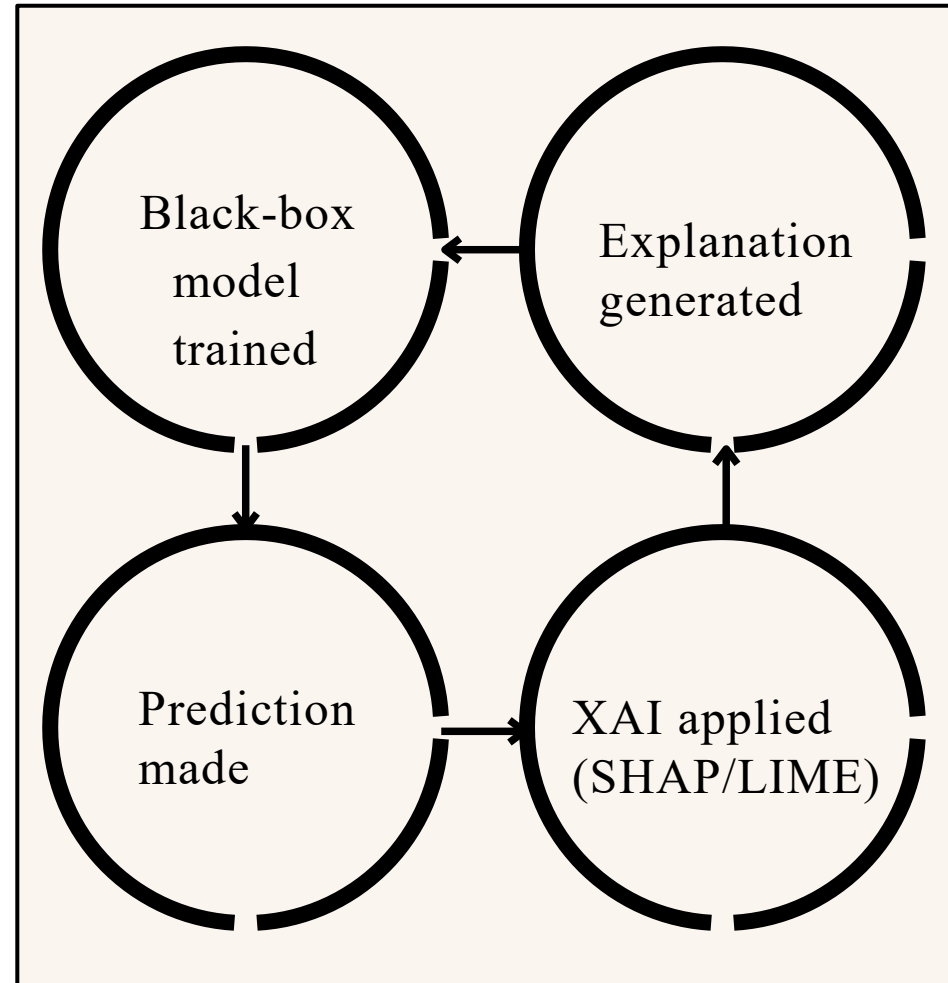
- | | |
|-------------|-----------|
| Local | linear |
| • surrogate | Model- |
| • agnostic | Instance- |
| • level | |

Problem Statement

- Most ML deployments in high-stakes fields lack transparent, reproducible explainability frameworks.
- Existing XAI research evaluates methods in isolation without multi-model, multi-dataset benchmarking.
- Explanation stability and consistency across re-training runs are rarely assessed or reported.
- Real-world data complexity (class imbalance, missing values, correlated features) is often ignored.



Problem Statement



Methodology

- 1 Literature Review**
Surveyed SHAP, LIME, and XAI benchmarking literature across healthcare, credit, and tabular data domains.
- 2 Dataset Preprocessing**
Three UCI datasets processed: median/mode imputation, one-hot encoding, z-score normalization, and SMOTE balancing.
- 3 Model Training**
Python-based simulations were developed using datasets to model sponsored search ranking behavior.
- 4 Explanation Generation**
SHAP (TreeExplainer/KernelExplainer) and LIME applied to generate global and local feature attributions.

Results & Implementation

Top Performer: XGBoost

XGBoost led with AUC-ROC 0.927 (Adult Income) and 0.891 (German Credit); RF led on Pima Diabetes at 0.846.

SHAP vs LIME Agreement

SHAP and LIME feature attributions showed mean Spearman correlation of 0.73 across all models and datasets.

SHAP Stability

GBM ($\tau=0.88$) and RF ($\tau=0.85$) showed highest explanation stability across 10 independent re-training seeds.

Python Stack

Implemented with scikit-learn 1.3, XGBoost 2.0, SHAP 0.42, LIME 0.2 on Python 3.10 with RTX 3060 GPU.

Key Findings

- Ensemble models (XGBoost, Random Forest) consistently outperform Logistic Regression and SVM across all three datasets.
- SHAP provides more stable and faithful feature attributions compared to LIME, especially near decision boundaries.
- Top predictive features aligned with domain expertise: glucose and BMI for diabetes; capital gain for income prediction.
- Explanation stability (Kendall tau > 0.8) should be treated as a primary evaluation criterion alongside accuracy.

Conclusion

Performance vs. Interpretability

Ensemble models yield optimal trade-off between predictive accuracy and consistent, faithful feature-level explanations.

Multi-Dataset Validation

Framework validated on UCI Adult Income, Pima Diabetes, and German Credit datasets spanning diverse domains.

Modular XAI Pipeline

XMLF combines four classifiers with SHAP and LIME in a plug-and-play, domain-independent pipeline for responsible AI.

Stability as Key Criterion

Explanation stability across re-training runs must be a primary model selection criterion.

Future Scope

- 1 Counterfactual Explanations**
Integrate counterfactual explanation algorithms (e.g., DICE, NICE) into the XMLF for contrast-based reasoning.
- 2 Algorithmic Fairness**
Extend the framework to include fairness evaluation metrics to detect and mitigate model bias across demographic groups.
- 3 Interactive Expert Review**
Develop an interactive UI for domain experts (clinicians, credit analysts) to review and challenge model explanations.
- 4 Image and Text Data**
Expand beyond tabular data to apply XMLF explainability techniques to image and natural language classification tasks.

Thank You